

DevOps Engineer

Step 1: Learn Linux & Networking

CareerCompass • Detailed Concept Notes • Research-backed learning guide

Concept & Learning Objective

DevOps engineers operate systems, so the operating system and network model must become practical skills. Linux fundamentals include filesystem layout, processes, users/groups, permissions, services, logs, packages, environment variables and SSH. Networking explains how a request travels from a client through DNS, routing, TCP/TLS and an application server.

The goal is diagnostic ability: when a service fails, you should be able to decide whether the problem is DNS, connectivity, TLS, application behavior, permissions, resources or configuration.

Core Concepts — Detailed Breakdown

- Linux filesystem and paths: /etc, /var, /home, /tmp, /proc and service configuration.
- Processes: PIDs, foreground/background jobs, signals, process states and resource usage.
- Permissions: users, groups, rwx bits, ownership, sudo and least privilege.
- Services/logs: systemd concepts, service status and log inspection.
- Shell: pipes, redirection, grep, find, environment variables and exit codes.
- Networking: IP addressing, subnets at a basic level, ports and protocols.
- DNS: names map to IP addresses; understand records, resolvers and TTL.
- TCP vs UDP: reliability/ordering vs lower-overhead datagrams.
- HTTP/HTTPS: request/response, TLS, status codes and headers.
- Troubleshooting: ping/curl/ss/dig/traceroute/ssh and log inspection.

What You Should Be Able To Do

- Create users and groups.
- Change file ownership and permissions.
- Start/stop and inspect a service.
- Use SSH.
- Resolve a hostname and inspect HTTP headers.
- Identify which process is listening on a port.

Hands-on Project / Practice

- Run a Linux VM. Install a web server, expose it on a known port, inspect its process/logs, diagnose a deliberate DNS or port mistake, and document the commands used.

Recommended References

- Linux Journey — <https://linuxjourney.com/>
- Linux Tutorial For Beginners — https://www.youtube.com/watch?v=_tCY-c-sPZc

Note: These notes are designed to teach the concepts and provide a practical learning path. Always prefer the linked official documentation for current APIs, SDK versions and platform-specific release requirements.